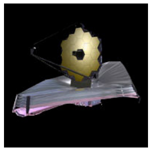

1572 - Mapping, resolving and penetrating into the dusty Spiderweb protocluster with unqiue Pa-beta imaging

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
PaB imaging of Spiderweb protocluster				
	1	NIRCam PaB imaging of Spiderweb	NIRCam Imaging	(1) SPIDERWEB

ABSTRACT

We propose to conduct a unique and very efficient Pa-beta imaging (1.4hrs science time!) of the best-studied galaxy cluster in formation, called "Spiderweb" protocluster, at z=2.16, with the NIRCam F405N filter. Our primary goal is to map the entire star formation activities across this starbursting protocluster, and within individual galaxies therein, by spatially resolving the dust-enshrouded star forming regions. We will unveil all the star forming activities down to a SFR of 3.5 solar masses per year, free from dust extinction, making a complete census of the protocluster (core). By mapping the SFR(Pa-beta)/SFR(UV) ratio within the galaxies in combination with the existing HST/ACS data, we will pinpoint the locations of

dusty star-forming regions within the young cluster galaxies, - for a subset of members even matching at a similar spatial resolution of 1kpc ALMA dust continuum imaging. Taking advantage of the multi-wavelength, ancillary data available for the target field, we will calibrate the different star-formation indicators such as H-alpha, Pa-beta, dust continuum and rest-UV to NIR SEDs and we will determine the star-formation mode of each member galaxy. Simultaneously for free, we will observe with F115W (rest-U) and F182M (rest-V) at the SW channel of NIRCam, to map the distribution of quiescent and star-forming regions (and stellar mass surface density) for >100 known cluster members (out of which 61 are z-spec confirmed). The proposed Pa-beta imaging - as this line falls luckily in F405N - is the most powerful, unbiased approach to reveal the complete picture of the stellar mass assembly process in this young protocluster environment at cosmic noon.

OBSERVING DESCRIPTION

We propose redshifted Pa-beta imaging with the NIRCam F405N filter of a galaxy protocluster at z=2.16. In addition, we request F410M broad-band imaging to subtract the continuum and thus obtain Pa-beta line emission. For the SW channel we choose F115W and F162M. The combination of the proposed broad-/medium-band filters (F115W, F182M, F410M) trace the rest-frame U, V , J-bands. We exploit a primary dither pattern of "FULLBOX 4TIGHT" with ROWSx2 (overlap=35%) and COLUMNx1 to map ~3'x6' FoV where HST/ACS (G475W, I814W) data exist. With the image quality of 0.14" of NIRCam at 4micron, we can spatially resolve the Pa-beta-based SF activity within the galaxies on ~1-kpc scale, which is comparable to the existing HST (rest-UV) data.

The requested observing times are 3780 sec (for F405N and F115W) and 1288 sec (for F410M and F182M), with which we can achieve 5-sigma point source sensitivity of 28.5 mag (F115W), 27.0 mag (F182M), 26.2 mag (F405N), and 27.3 mag (F410M). Accordingly, we exploit the read-out patterns of "SHALLOW4" (BRIGHT1) with Ngroups=9 (8), NINT=1 (1), and NEXP=8 (8) for F405N (and F410M), respectively.

Proposal 1572 - Targets - Mapping, resolving and penetrating into the dusty Spiderweb protocluster with unqiue Pa-beta imaging

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SPIDERWEB	RA: 11 40 47.8000 (175.1991667d) Dec: -26 29 21.00 (-26.48917d) Equinox: J2000		

Comments:
Category=Clusters of Galaxies
Description=[High-redshift clusters]

Observation

Proposal 1572, Observation 1: NIRCam PaB imaging of Spiderweb

Wed Mar 31 03:00:59 GMT 2021

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SPIDERWEB	RA: 11 40 47.8000 (175.1991667d) Dec: -26 29 21.00 (-26.48917d) Equinox: J2000		

Comments:
Category=Clusters of Galaxies
Description=[High-redshift clusters]

Template

Module	Subarray
ALL	FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift	Column shift	Tile Order
2	1	35.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	4TIGHT	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F115W	F405N+F444W	SHALLOW4	9	1	4	4	1889.672	
2	F182M	F410M	BRIGHT1	8	1	4	4	644.206	

Special Requirements

Group Visits within 53.0 Days
Visits Same PA